

# Sankalp Singh Bais

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## Edge Platform / DevOps Engineer - Linux, Containers, Real-Time Systems

### TECHNICAL SKILLS

**Languages:** Python, Bash

**Linux & Networking:** systemd, crontab, nmcli, RTSP, WireGuard, Headscale

**Containers & Deployment:** Docker, GitLab CI/CD, Azure Container Apps, Azure Container Registry, Dokploy, Proxmox, Ansible

**Backend & Observability:** Flask, Celery, Redis, REST APIs, Prometheus, Grafana, Loki, Node Exporter

**Edge & Inference:** Raspberry Pi 4/5, Jetson Nano, Hailo HAT (SDK), PoE Cameras, Ublox GPS, ONNX Runtime, LPRNet, YOLOv8

### EXPERIENCE

#### Technical Trainee - Edge Platform & DevOps

June 2024 - Present

Containova Systems LLP

Navi Mumbai, India

- **Owned a production edge platform** deployed across **26+ live devices**, responsible for **architecture, CI/CD, deployments, monitoring, upgrades, and incident response**
- **Designed and operated GitLab CI/CD pipelines** to **build, test, containerize, and deploy** services to **Azure Container Apps** via **Azure Container Registry**
- **Migrated workloads from Azure App Services to Azure Container Apps**, achieving **60–70% cloud cost reduction** and improving **rollout and rollback reliability**
- **Established cloud-on-prem deployment parity** by deploying **Dockerized services and databases on-prem using Dokploy**, while **retaining Azure deployments** for existing cloud clients
- **Re-architected backend services** from **threaded Flask to Celery + Redis**, enabling **reliable concurrency** for **multi-device workloads**
- **Implemented centralized observability** using **Prometheus, Grafana, and Loki** to monitor **uptime, inference latency, device health, and network stability** across the edge fleet
- **Led edge runtime migration** from **Jetson Nano to Raspberry Pi 5 + Hailo HAT**, benchmarking and upgrading inference from **YOLOv7 + EasyOCR to YOLOv8 + LPRNet**, achieving **~0.07s/frame** latency with Hailo and **~0.3s/frame** on CPU using ONNX Runtime
- **Acted as primary production support engineer**, owning **on-call incident response** for outages, **device failures**, and **critical client escalations**

#### Hardware and Software Intern

July 2023 - June 2024

Containova Systems LLP

Navi Mumbai, India

- **Designed the initial hardware-software architecture** for an **industrial container detection system**, which later evolved into the **production edge platform**
- **Built a dual-unit edge prototype** with **Jetson Nano for inference** and **Raspberry Pi + Ublox GPS for positioning and control**, integrating compute, vision, and location data
- **Implemented RTSP pipelines** from **PoE cameras** and automated **video capture workflows** to support **dataset collection and system validation**
- **Implemented GPS ingestion via a private protocol**, parsing and serving **location data over HTTP** for downstream services

### PROJECTS

#### System Monitoring Dashboard (Personal Initiative)

August 2025

- Designed and built a **centralized monitoring stack** using **Prometheus, Grafana, and Loki** as a personal initiative, later **deployed across 26+ production edge devices**
- Used to **monitor uptime, resource utilization, and inference latency** across multiple client environments.
- Fully **VPN-isolated** with zero public exposure, following production-grade security practices.

#### NORS Automata v1.0 Home Automation

March 2023

- Built a multi-modal automation system using **Raspberry Pi, Python, Django, WebSockets, OpenCV**.
- Implemented real-time state synchronization using **ASGI/WebSockets**.
- Integrated voice and gesture control using **Google STT** and **Hugging Face models**.